

WHIS AI — CORE MISSION, ARCHITECTURE & STRATEGY

■ WHIS AI — CORE MISSION, ARCHITECTURE & STRATEGY (CORE.md)

1. THE CORE AIM

The core aim of **Whis AI** is to protect India's next **300 million first-time digital banking users** in Tier-2, Tier-3, and rural areas from devastating financial losses and psychological harassment caused by cyber fraud.

Core Mission Objectives:

- **Prevent Financial Extortion:** Stop "Digital Arrest" impersonation calls from fake Police, CBI, or Customs officers before the victim engages.
- **Neutralize Phishing & Malware:** Automatically detect fake bank KYC update links, electricity bill disconnection traps, and malicious loan app .apk downloads in SMS and notifications.
- **Eliminate UPI Misconceptions:** Enforce the universal rule: **"UPI PIN is ONLY for SENDING money, NEVER for RECEIVING."**
- **De-Escalate Psychological Panic:** Provide clear, colloquial Hinglish guidance (**"GHABRAO MAT"**) and instantly bridge victims to India's **1930 Cyber Crime Helpline** and their family.

2. HOW WE ACHIEVE THE AIM

Whis AI achieves its mission through a non-intrusive, hybrid, privacy-first mobile architecture:

1. **Pre-Emptive Screen Interruption:** Instead of logging scams after the loss occurs, Whis screens incoming callers in **~46ms** using Groq Llama 3.3 70B AI. If a Digital Arrest threat is detected, Whis interrupts the user's screen with a full-screen `RedAlertActivity` before the call is answered.
2. **5-Layer Hybrid Detection Pipeline:** Combines zero-latency local checks (DLT header whitelisting, keyword rules, on-device ML, URL expansion) with ultra-fast cloud LLM reasoning. Messages escalate to cloud AI only when local layers are inconclusive, preserving speed and privacy.
3. **Automated Family Emergency SOS Network:** Dispatches automated emergency SMS alerts to **2 Emergency Contacts** configured by the user, creating an immediate family safety net.
4. **DPDP Act 2023 Compliance:** Converts SMS message bodies into **SHA-256 hashes** before storing them in local SQLite databases, guaranteeing zero plaintext message persistence.

3. STRATEGIC EXECUTION PLAN

[Phase 1: Interception] ■■■ [Phase 2: Hybrid Detection] ■■■ [Phase 3: AI Intent Reasoning] ■■■ [Phase 4: Action & Intervention]

- **Phase 1 (Interception):** Intercept incoming voice calls via `PhoneStateReceiver` and SMS/App notifications via `WhisNotificationListenerService`.
- **Phase 2 (Hybrid Detection):** Run local DLT whitelisting (Layer 1), keyword scam taxonomy matching (Layer 2), on-device TFLite classification (Layer 3), and `UrLExpander` link analysis (Layer 4).
- **Phase 3 (AI Intent Reasoning):** Dispatch ambiguous payloads to **Groq Llama 3.3 70B** (`gsk_...`) for sub-second semantic analysis, with automatic failover to **Groq 8B (30k TPM)** or **Google Gemini Flash**.
- **Phase 4 (Action & Intervention):** Trigger `RedAlertActivity` full-screen overlay for scam calls, post heads-up warning notifications for phishing SMS, update the Room database, and alert **2 Emergency Contacts**.

4. WHAT IS THE MIND MAP (SYSTEM VISUALIZATION)

graph TD

A["■ INCOMING THREAT VECTORS"] --> B1["■ Incoming Phone Call
(PhoneStateReceiver)"]

A --> B2["■ Incoming SMS Message
(SmsReceiver)"]

A --> B3["■ App Notification (WhatsApp/Bank)
(WhisNotificationListenerService)"]

A --> B4["■ On-Demand User Chat
(Ask Whis AI)"]

%% Known Contact Check

B1 --> C1["Is Caller in Address Book?"]

C1 -- Yes --> D1["■ SAFE: Marked as 'Trusted' (0ms)
Saved to CallHistoryStore"]

C1 -- No (Unknown) --> C2["CallFilterService"]

%% 5-Layer Engine

B2 & B3 --> E["5-LAYER HYBRID DETECTION ENGINE
(WeightedScoreEngine)"]

E --> L1["Layer 1: TRAI DLT Header Verification
(-HDFCBK, -SBIBNK Whitelist)"]

L1 --> L2["Layer 2: Heuristic Scam Rule Engine
(KYC, Blocked, PAN, UPI, Police, CBI)"]

L2 --> L3["Layer 3: Local TFLite Token Classifier"]

L3 --> L4["Layer 4: URL & APK Expander
(Follows bit.ly redirect chains to detect .apk links)"]

L4 --> L5["Layer 5: Multi-Provider Cloud LLM Fallback"]

%% Multi-Provider AI Routing

C2 & L5 & B4 --> F["MULTI-PROVIDER AI ROUTER
(GeminiAgentClient)"]

F -- Primary (gsk_...) --> G1["Groq Llama 3.3 70B Versatile
(~46ms Ultra-Fast Speed)"]

G1 -- 429 Rate Limit --> G2["Groq Llama 3.1 8b Instant
(30,000 Tokens/Min Limit)"]

F -- Secondary (AlzaSy...) --> G3["Google Gemini 1.5 / 2.0 Flash"]

%% Action & Intervention System

G1 & G2 & G3 --> H["AI Threat Verdict"]

H -- SCAM Call / Digital Arrest --> I1["■ RED ALERT SCREEN (RedAlertActivity)
• Full Screen Overlay (USE_FULL_SCREEN_INTENT)
• 'GHABRAO MAT' De-escalation
• 1-Tap 1930 Cyber Crime Helpline
• Auto-SMS SOS to 2 Emergency Contacts"]

H -- SCAM / Phishing SMS --> I2["■ WARNING NOTIFICATION & BADGE
• Heads-Up Red Alert Notification
• Stored in LocalMsgDatabase (SHA-256 Hash)
• Appears in Messages & Home Feed"]

H -- SAFE Transactional --> I3["■ TRUSTED / LIKELY SAFE
• 0/100 Risk Score
• Green Badge in Feed"]

%% Data & Storage Layer

I1 & I2 & I3 --> J["DATA & STORAGE LAYER"]

J --> K1["LocalMsgDatabase (Room SQLite)
SHA-256 Body Hashes (DPDP Act Compliant)"]

J --> K2["CallHistoryStore
SharedPreferences LIFO Persistent Store"]

J --> K3["EmergencyContactStore
Stores Contact 1 & Contact 2 SOS Details"]

%% UI Dashboard Layer

K1 & K2 --> L["■ USER INTERFACE DASHBOARD"]

L --> M1["■ Home Tab: Unified Activity Feed (Calls + SMS sorted by timestamp)"]

L --> M2["■ Calls Tab: Screened Calls Feed & Blocked List"]

L --> M3["■ Messages Tab: Screened SMS & Alert Feed"]

L --> M4["■ Settings Tab: Permission Controls & 2 Emergency Contacts Editor"]

5. WHICH FEATURES ARE USED AND HOW THEY HELP

6. KEY DIFFERENTIATORS

- **Non-Intrusive Background Operation:** Whis does **not** demand becoming the default SMS handler, lowering adoption friction.
- **Resilient AI Architecture:** Dual Groq/Gemini routing eliminates single-point-of-failure rate limit crashes.

- **Interactive Cyber Education:** Includes the **Learn Tab** (LearnRepository) with interactive scam chapters (Digital Arrest, Phishing, UPI Rules, APK Malware) and RBI Zero-Liability refund guidelines.

7. EXECUTIVE SUMMARY

Whis AI is a complete, production-grade Android application (targetSdk 36, minSdk 24) that combines ultra-fast AI intelligence, privacy-first local hashing, full-screen emergency de-escalation, and dual-contact SOS alerts. It provides the next 300 million digital banking citizens of India with an unbreachable, non-intrusive cyber safety shield.

- **Repository:** <https://github.com/govind-ma/whis-ai>

- **License:** MIT License